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To the Editor, *Scientific Reports*

Dear Dr Zoete and Ms Agnihotri,

Please find the second revision of our manuscript, now titled "BioInteract: Interpretable Drug–Target Interaction Prediction via Residue-Level Cross-Attention with Protein-Language and Physicochemical Priors" (Submission ID 56f53c41-a86e-4948-8cfd-a3c11b83fd1b), for reconsideration in *Scientific Reports*. The revision comprises a clean manuscript, a marked manuscript in which new and substantively revised material is shown in red, Supporting Information, and a point-by-point response.

Your decision letter asked us to address methodological novelty, the need for an appropriate ESM-2 baseline, the reproducibility of the BindingDB filtering procedure, comparison with recent models, the claimed biological meaning of the attention maps, the purpose of the mutation experiments, and the remaining presentation issues. We have run new experiments for the first four of these rather than only rewriting text.

The most consequential addition is a matched-protocol benchmark in which BioInteract and re-implemented baselines are trained by a single harness under one partition, seed, optimiser, schedule, loss and threshold rule. The main text keeps a minimal three-model ladder that changes one component at a time and whose middle row consumes exactly the frozen ESM-2 representation BioInteract uses, which provides the controlled comparison Reviewer 2 requested; three further architecture families evaluated in our own setting are reported in the Supporting Information. The main comparison table no longer contains any value quoted from another publication. Within the controlled ladder each component earns its place, and cross-attention contributes most under the entity-held-out protocols (0.201 AUPRC over concatenation on Target-ID-held-out, against 0.067 on the random split). The random split itself separates the tested architectures only weakly, so we rest no conclusion on it. The Abstract, Discussion and Conclusion have been rewritten so that the claims match the controlled evidence. The paper is now positioned as an audited, reproducible and applicability-domain-aware reference point for the Davis binary classification task rather than as a new architectural principle.

We have also reported the BindingDB curation as a complete ordered filtering waterfall, which shows that 97.2 % of the rows in the pre-specified snapshot report no K_d at all, and we applied the identical pipeline to the complete BindingDB release to demonstrate that the criteria scale as expected. A new applicability-domain analysis explains the limited external transfer quantitatively, and the released web server now reports the same two nearest-neighbour scores for every user-submitted pair, so that a user is warned when a query lies in the regime where our external evaluation showed chance-level ranking. While preparing the revised web-server figure we found and corrected a mislabelled example in the previous version, which is described in the response letter.

Every presentation change requested by Reviewer 3 has been implemented. Three tables and one figure were deleted, one figure was merged, and one was replaced by a more informative version, so the manuscript contains fewer floats than before despite the three new analyses.

All code, checkpoints, curation and audit scripts, and the analyses added in this round are available through the project GitHub repository and the immutable Zenodo release cited in the Code Availability section.

Thank you for considering our revision.

Sincerely,
Zhi Huang, on behalf of all authors